BM3122 | MEDICAL IMAGING

DOCUMENT A

Past paper pattern analysis

Evidence-led analysis of the 2023, 2024 and 2025 examinations, mapped to the supplied course material. Predictions are preparation priorities, not claims about the 2026 paper.

1. Source audit and interpretation

All 13 named PDFs were considered: six lecture decks, the two-page syllabus, four past-paper files and two assignments. There are **three independent examination years**, not four. The 2023 files reproduce the same six-page exam; one 14-page file interleaves eight pages of handwritten solutions. The file named “with ans” is a six-page typeset paper and contains no separate worked answer pages. Do not count either duplicate as another observation.

Source	Pages / status	Use
BM3122-S1 / S2 / S3 / S4 / S05 / S06	74 / 70 / 77 / 108 / 40 / 40 = 409 lecture pages	Main content and slide references; S05 title date is 2025, other supplied decks 2026.
BM3122 Medical Imaging	2 pages; intake 2020 onwards	Official module outline: six syllabus groups, 50% CA / 50% written exam.
BM3122_Nov2023	14 PDF pages; exam printed pp.1-6 at PDF pp.1,2,3,6,8,11	2023, Batch 20; handwritten answers are unverified student work.
BM3122_Nov2023_with ans	6 PDF pages	Duplicate 2023 paper, clean typeset question source.
BM3122_Nov2024	6 PDF pages	2024, Batch 21; Figure Q2 inspected visually.
Medical Imaging - Batch22	6 scanned pages, visually inspected	Cover explicitly says November 2025, Batch 22; Figures Q1/Q3 inspected.
230507R_A1 / 230507R_A2 (1)	10 / 9 PDF pages	Student assignments on thermal/OCT. Not lecturer slides or an official marking scheme.
Additional local materials checked	Final_exam_guide.png; Last day que discussion_Thisura.pdf (6 pages)	Guide is a secondary trend summary, not an authenticated lecturer instruction. Discussion repeats decay, population, Larmor, US timing, reflection budget and DSA. No verified exclusions.

References: S1, S2, S3, S4, S05 and S06 mean the corresponding named lecture PDFs; numbers are PDF page numbers, matching visible numbered slides. A1/A2 mean the supplied thermal/OCT assignments and always use PDF page numbers, including cover and contents. Printed past-paper question numbers are used even in the interleaved 2023 file. Text extraction exposes an unrelated 2005 memorandum layer on some cover pages; it is not exam content or an instruction for this analysis.

2. Verified examination structure

Year / batch	Format	Question blocks / marks
2023 / 20	2 hours; 100 marks; 4 compulsory questions; 25 each; 6 printed pages	Q1 Intro/ADC 15 + NM 10; Q2 X-ray 11 + CT 14; Q3 MRI 25; Q4 US 17 + thermal 8
2024 / 21	2 hours; 100 marks; 4 compulsory questions; 25 each; 6 printed pages	Q1 Image quality/intro 10 + NM 15; Q2 X-ray 12 + CT 13; Q3 MRI 25; Q4 US 19 + OCT 6

Year / batch	Format	Question blocks / marks
2025 / 22	2 hours; 100 marks; 4 compulsory questions; 25 each; 6 printed pages	Q1 Image quality/intro 10 + NM 15; Q2 X-ray 14 + CT 11; Q3 MRI 25; Q4 US 17 + OCT 8

All covers state closed book, answer ALL, equal marks, derivations only when explicitly asked, and an attached information sheet. They permit reasonable stated assumptions for missing data/ambiguous wording. These are **observed historical instructions**; no 2026 cover or lecturer notice was supplied. The models adopt the stable two-hour/100-mark structure and supply their own corrected data sheet. Allow roughly 30 minutes per 25-mark question (1.2 minutes per mark).

Subquestion structure is not identical: 2023 uses 7/7/7/7 parts across Q1-Q4; 2024 uses 7/6/7/8; 2025 uses 7/8/9/8. MRI has 25 marks every year but shifts substantially in content. Small optical/thermal parts are compulsory components, not optional alternatives.

3. Complete subquestion classification

One row per printed lowest-level subquestion. Types are analyst classifications of the dominant demand; mixed questions may also require explanation or calculation. No missing scan content or marks were inferred. Figure-dependent items are explicitly identified.

2023 question inventory

Year / question	Topic	Subtopic / tested concept	Type	Marks	Lecture / source
2023 1(a)i	Introduction	3 ionizing + 3 non-ionizing modalities	Recall	3	S1 6,29
2023 1(a)ii	Introduction	Pixel meanings for named modalities	Explain	4	S1 31
2023 1(b)i	Image acquisition	Three ADC specifications	Recall	3	S1 62
2023 1(b)ii	Image acquisition	Quantization error	Explain	2	S1 65-66
2023 1(b)iii	Image acquisition	Two ways to reduce relative quantization error	Explain	3	S1 65
2023 1(c)i	Nuclear medicine	Activity purchase 21 h before use; 1.7 h half-life	Numerical	5	S06 6-7
2023 1(c)ii	Nuclear medicine	Equal activity: 100 GBq In / 20 GBq Tc	Numerical	5	S06 6-7
2023 2(a)i	X-ray	DSA pre-contrast intensity	Equation	2	S2 27-28,48-49
2023 2(a)ii	X-ray	DSA post-contrast intensity	Equation	2	S2 27-28,48-49
2023 2(a)iii	X-ray	Why direct subtraction retains tissue	Derivation	3	S2 27-28,48-49
2023 2(a)iv	X-ray	Log subtraction removes static tissue	Derivation	4	S2 27-28,48-49
2023 2(b)i	CT	2D Radon transform of unit disk	Derivation	4	S3 4,11-16; disk result not explicitly stated
2023 2(b)ii	CT	Three features each: generations 3 and 4	Compare	6	S3 37-38
2023 2(b)iii	CT	Four routine QA tests	Recall	4	QA not explicit in S3; HU/resolution context S3 31-32
2023 3(a)i	MRI	Three limitations	Recall	3	S4 5
2023 3(a)ii	MRI	Main scanner components	Equipment	6	S4 93-101

Year / question	Topic	Subtopic / tested concept	Type	Marks	Lecture / source
2023 3(a)iii	MRI	Three imaging magnet requirements	Recall	3	S4 94
2023 3(b)i	MRI	Net proton population percentage, 3 T at 24 C	Numerical	5	S4 12-17
2023 3(b)ii	MRI	Larmor frequency range, central 3 mm slab	Numerical	3	S4 15,48-54
2023 3(b)iii	MRI	RF bandwidth; gradient 0.5 mT/cm	Numerical	3	S4 52-54
2023 3(b)iv	MRI	Two phantom-tested system parameters	Recall	2	S4 52-54,94,98; tests not explicitly listed
2023 4(a)i	Ultrasound	Three advantages + three disadvantages	Compare	3	S05 3-4
2023 4(a)ii	Ultrasound	Tissue interactions and image consequences	Explain	6	S05 13-16
2023 4(a)iii	Ultrasound	20 cm; choose 1, 2 or 5 MHz; 100 dB budget	Numerical	5	S05 16,28-29
2023 4(a)iv	Ultrasound	Maximum scan lines at 100 Hz	Numerical	3	S05 6-7,17,19; timing application
2023 4(b)i	Thermal	Operation principle	Explain	4	A1 PDF 3-5; syllabus p.1
2023 4(b)ii	Thermal	Four medical applications	Recall	2	A1 PDF 5-7
2023 4(b)iii	Thermal	Four advantages	Recall	2	A1 PDF 3,5-7

2024 question inventory

Year / question	Topic	Subtopic / tested concept	Type	Marks	Lecture / source
2024 1(a)i	Introduction	3 ionizing + 3 non-ionizing modalities	Recall	3	S1 6,29
2024 1(a)ii	Image quality	One spatial-resolution factor for each modality	Explain	3	S1 39-48; modality-specific lectures
2024 1(a)iii	Image quality	Derive Gaussian FWHM	Derivation	4	S1 41-44
2024 1(b)i	Nuclear medicine	Four SPECT radiotracer properties	Recall	4	S06 9-10
2024 1(b)ii	Nuclear medicine	Three reasons PET has higher SNR	Explain	3	S06 25-27
2024 1(b)iii	Nuclear medicine	Main PET limitation	Recall	2	S06 28-29
2024 1(b)iv	Nuclear medicine	Equal in-body activities; physical + biological half-lives	Numerical	6	S06 6-8
2024 2(a)i	X-ray	Two contrast-agent classes + applications	Recall	4	S2 43-47
2024 2(a)ii	X-ray	Three interaction mechanisms	Recall	3	S2 17-25
2024 2(a)iii	X-ray	HVL bar phantom image contrast	Numerical	5	S2 27-31; S1 55
2024 2(b)i	CT	Three advantages over projection radiography	Compare	3	S3 2-3,31-32
2024 2(b)ii	CT	Central section theorem	Explain	4	S3 15-16
2024 2(b)iii	CT	3x3 ART reconstruction from Figure Q2	Numerical	6	S3 26-28
2024 3(a)i	MRI	Three intensity factors	Recall	3	S4 3,26,75
2024 3(a)ii	MRI	Four advantages	Recall	2	S4 4
2024 3(b)i	MRI	Basic GE timing diagram	Diagram	5	S4 63-65

Year / question	Topic	Subtopic / tested concept	Type	Marks	Lecture / source
2024 3(b)ii	MRI	Modify GE for multislice without longer scan	Diagram	4	S4 72-74
2024 3(b)iii	MRI	Tip angle meaning	Explain	2	S4 20-22
2024 3(b)iv	MRI	Ernst angle for TR=1.5 T1	Numerical	2	S4 78
2024 3(b)v	MRI	SNR loss with 90 degrees instead of optimum	Numerical	7	S4 78
2024 4(a)i	Ultrasound	Four advantages	Recall	2	S05 3
2024 4(a)ii	Ultrasound	Three artifacts + causes	Explain	6	S05 35-38
2024 4(a)iii	Ultrasound	Two methods to reduce an artifact	Explain	3	S05 32-35
2024 4(a)iv	Ultrasound	256 elements in non-overlapping groups of 8: lines	Numerical	2	S05 23-24
2024 4(a)v	Ultrasound	Frame rate at 50 cm, c=1540 m/s	Numerical	4	S05 6-7,17,19; timing application
2024 4(a)vi	Ultrasound	How to increase frame rate	Explain	2	S05 19,23-24; timing application
2024 4(b)i	OCT	Two applications	Recall	2	A2 PDF 6-7
2024 4(b)ii	OCT	Operation principle	Explain	4	A2 PDF 3-5

2025 question inventory

Year / question	Topic	Subtopic / tested concept	Type	Marks	Lecture / source
2025 1(a)i	Image quality	Spatial resolution definition	Recall	1	S1 39
2025 1(a)ii	Image quality	Four object/PSF convolution sketches, Figure Q1	Diagram	6	S1 45-48
2025 1(a)iii	Image quality	Object most affected by PSF; justify	Explain	3	S1 45-48
2025 1(b)i	Nuclear medicine	Two advantages + two disadvantages	Compare	4	S06 2-4
2025 1(b)ii	Nuclear medicine	PET annihilation coincidence detection	Explain	3	S06 25,29-34
2025 1(b)iii	Nuclear medicine	Two hybrid PET/CT advantages	Recall	2	S06 27
2025 1(b)iv	Nuclear medicine	Equal initial atom numbers; Tc counts below 20% of total	Numerical	6	S06 6-7; equal detection efficiency unstated
2025 2(a)i	X-ray	Planar system main components and functions	Equipment	6	S2 32-42
2025 2(a)ii	X-ray	Two ways to reduce effective focal spot	Recall	2	S2 9-10
2025 2(a)iii	X-ray	Effects of smaller focal spot on image	Explain	3	S2 9-11,50
2025 2(a)iv	X-ray	Transmission through 1 cm at mu=0.3 /cm	Numerical	3	S2 27-28
2025 2(b)i	CT	Why define Hounsfield units	Explain	2	S3 31-32
2025 2(b)ii	CT	Linear calibration from air=-1100 and water=10	Numerical	4	S3 31-32; calibration application
2025 2(c)i	CT	Analytic vs iterative reconstruction	Compare	3	S3 14
2025 2(c)ii	CT	Example of each reconstruction method	Recall	2	S3 15-29

Year / question	Topic	Subtopic / tested concept	Type	Marks	Lecture / source
2025 3(a)i	MRI	Three intensity factors	Recall	3	S4 3,26,75
2025 3(a)ii	MRI	RF range for 3 mm slice; Gz=0.3 mT/cm	Numerical	4	S4 15,52-54; B0 not supplied
2025 3(a)iii	MRI	Two ways to reduce slice thickness	Explain	2	S4 52-54
2025 3(b)i	MRI	Timing/purpose of each GE gradient in Figure Q3	Explain	3	S4 52-65
2025 3(b)ii	MRI	Compress pulse sequence without changing image	Diagram	3	S4 63
2025 3(b)iii	MRI	256x256 scan time, TR=4500 ms	Numerical	2	S4 61-62,71
2025 3(b)iv	MRI	Multislice modification in same scan time	Diagram	4	S4 72-74
2025 3(b)v	MRI	Spin echo advantage over GE	Compare	2	S4 79-81
2025 3(b)vi	MRI	Mechanism: 180 degree refocusing	Explain	2	S4 79-81
2025 4(a)i	Ultrasound	Air gap physical problem + remedy	Explain	3	S05 13-15
2025 4(a)ii	Ultrasound	Speckle explanation	Explain	3	S05 35
2025 4(a)iii	Ultrasound	Resolution/penetration trade-off + two responses	Explain	4	S05 28-29
2025 4(a)iv	Ultrasound	2 MHz, 80 dB budget: maximum depth	Numerical	4	S05 16,28
2025 4(a)v	Ultrasound	256x256 B-mode frame rate at that depth	Numerical	3	S05 6-7,17,19; 256 sequential lines assumed
2025 4(b)i	OCT	Pixel represents what?	Explain	2	A2 PDF 3,5
2025 4(b)ii	OCT	One advantage + disadvantage vs ultrasound	Compare	3	A2 PDF 3,5,8
2025 4(b)iii	OCT	One major application explained	Explain	3	A2 PDF 6-7

4. Topic frequency and marks

Topic	2023 marks	2024 marks	2025 marks	Years present
Introduction	7	3	0	2/3
Image quality	0	7	10	2/3
Image acquisition	8	0	0	1/3
Nuclear medicine	10	15	15	3/3
X-ray	11	12	14	3/3
CT	14	13	11	3/3
MRI	25	25	25	3/3
Ultrasound	17	19	17	3/3
OCT	0	6	8	2/3
Thermal	8	0	0	1/3

The first three rows divide the introductory lecture by the dominant demand. Their combined marks are 15, 10 and 10. Nuclear medicine, X-ray, CT, MRI and ultrasound appear in every independent year. OCT appears twice

and thermal once. “Not seen” means not directly assessed in these three papers; it does not establish exclusion. No evidence identifies a newly introduced topic; later appearance can reflect rotation rather than a syllabus change.

Dominant task category	2023	2024	2025	Typical individual allocation
Recall	22	25	10	1-4
Explain	19	27	33	2-6
Compare	9	3	12	2-6
Equipment	6	0	6	6-6
Equation	4	0	0	2-2
Derivation	11	4	0	3-4
Diagram	0	9	13	3-6
Numerical	29	32	26	2-7

5. Questioning style and changes

Introductions and progression

Questions open with a short modality description, then use “state”, “name”, “briefly explain”, “calculate”, “show” and “sketch”. Many move from recall to explanation and a linked calculation. Q4 repeatedly introduces ultrasound, then ends with OCT or thermal imaging.

Recurring frameworks

Nuclear medicine repeatedly changes the decay initial conditions; equal activity is not the same as equal atom number. Ultrasound repeatedly connects propagation losses to highest usable frequency/depth and then to echo travel time/frame rate. MRI links spatial encoding to gradients/RF and to practical acquisition time.

Diagrams and derivations

2023 asks DSA log algebra and disk Radon transform. 2024 asks Gaussian FWHM, ART, GE and multislice diagrams. 2025 uses convolution sketches and GE compression/interleaving. The figures are functional engineering representations, not decorative drawings.

Expected depth

At 2-3 marks, give distinct concise points or one equation with an interpretation. At 4-6 marks, show a mechanism, labeled sketch or all numerical steps and units. The 7-mark 2024 SNR comparison needs substitution, a ratio and a percentage loss; a final number alone is insufficient.

Recent change: justified but limited

Spatial-resolution emphasis increases from ADC in 2023 to LSF/FWHM in 2024 and PSF/convolution in 2025. Recent MRI emphasizes sequence timing rather than hardware. Q2 rotates among DSA/generations, central theorem/ART and focal spot/HU. There is no monotonic increase in mathematical difficulty: 2023 already

includes two derivations and 2024 contains a long GRE calculation.

Clinical and cross-topic demands

Applications mostly require examples and a physical explanation, particularly OCT/thermal and hybrid PET/CT. There is no separate case-diagnosis question in these papers. Links across lectures include common reconstruction for CT/PET/SPECT, resolution and pixel signals, and detector/noise trade-offs.

6. Complete syllabus coverage map

Topic / subtopic	Slide reference	Exact historical evidence	Frequency	Understanding / equations / diagrams	Preparation priority
Modalities, physical signals, structural/function	S1 3-31	2023 1(a); 2024 1(a)i	2/3	Energy, ionization, pixel meaning; MRI/US pixels are sequence/processing dependent	High
Spatial resolution; LSF, FWHM	S1 39-44	2024 1(a)ii-iii; 2025 1(a)i	2/3	Thin-line phantom, half-height Gaussian width	High
PSF, convolution and MTF	S1 45-48	2025 1(a)ii-iii	1/3	I=O*h; anisotropy; fine features blur; MTF magnitude	High
SNR, averaging, photon statistics	S1 50-53	2024 1(b)ii (PET context)	1/3 direct context	SNR scales as sqrt(counts) or sqrt(independent repeats)	Medium
Contrast, CNR, partial volume	S1 55-56	2024 2(a)iii (contrast)	1/3	Declare contrast definition; CNR=(SA-SB)/sigma	Medium
Artifacts	S1 58-59	2024 4(a)ii-iii; 2025 4(a)ii	2/3 US	Physical mechanism distinct from random noise	High for US
ADC, dynamic range, quantization	S1 62-66	2023 1(b)i-iii	1/3	n bits; step Vrange/2^n; gain without clipping	Medium
Sampling, oversampling, filtering	S1 67-74	No direct subquestion	0/3	Nyquist; anti-alias filter; fourfold oversampling per bit under assumptions	Coverage
X-ray tube, focal spot, heel, spectrum	S2 7-15	2025 2(a)i-iii	1/3	f=F sin(theta); heat/coverage; kVp/mAs; line+continuum spectrum	High
Photon interactions	S2 16-26	2024 2(a)ii	1/3	Photoelectric vs Compton vs Rayleigh; pair threshold	Medium
Attenuation, mass mu, HVL	S2 27-31	2023 2(a); 2024 2(a)iii; 2025 2(a)iv	3/3	Exponential law and sum of line integrals; HVL=ln2/mu	High
System, collimator, grid, CR/DR	S2 32-42	2025 2(a)i	1/3	Grid contrast/dose trade-off; scintillator vs direct charge conversion	High / coverage
Contrast agents and DSA	S2 43-49	2023 2(a)i-iv; 2024 2(a)i	2/3	Barium vs iodine; log subtraction; matched motion/exposure	High
Mammography, fluoroscopy	S2 50-53	No direct subquestion	0/3	Small focus, compression; dynamic acquisition and dose	Coverage
Dose and protection	S2 54-59	No direct subquestion	0/3	Gy vs Sv; tissue weighting; time/distance/shielding	Coverage
X-ray image examples	S2 60-70	No direct image-identification subquestion	0/3	Relate attenuation/contrast to anatomy; examples, not diagnoses to memorize	Coverage
CT projections, Radon, sinogram	S3 2-13	2023 2(b)i	1/3	-ln(I/I0); disk chord length; projection angle vs detector coordinate	High

Topic / subtopic	Slide reference	Exact historical evidence	Frequency	Understanding / equations / diagrams	Preparation priority
Central section theorem, Fourier reconstruction	S3 14-16	2024 2(b)ii; 2025 2(c)	2/3	Projection FT is radial slice of object 2D FT	High
Backprojection, FBP, ART, MLEM	S3 17-29	2024 2(b)iii; 2025 2(c)	2/3	Ramp filter; ray residual distribution; iterative forward projection	High
HU and calibration	S3 31-32	2025 2(b)	1/3	Water=0, air approximately -1000; linear calibration	High
CT generations, helical, multislice, dual energy	S3 35-43	2023 2(b)ii	1/3 generations	Motion geometry; pitch convention; detector rows; dual source vs dual energy	Medium / coverage
Special reconstruction, detectors	S3 45-54	2025 2(a)i only shared radiography hardware	0/3 CT-specific	Rebinning, interpolation, thin/thick slices; gantry signal path	Coverage
CT artifacts, applications, dose	S3 56-77	2023 QA adjacent, not direct artifact question	0/3 direct	Ring/streak/cupping; motion; clinical context; dose trade-offs	Coverage
CT QA	No explicit QA slide; S3 31-32 context	2023 2(b)iii	1/3	HU accuracy, uniformity, resolution, thickness; confirm local QA teaching	Medium
NMR, populations, Larmor, RF, detection	S4 7-28	2023 3(b)i-iii; 2024 3(b)iii	2/3	Net population difference, Hz vs rad/s, tip angle, Faraday induction	High
Relaxation, chemical shift, FID	S4 29-43	2024 intensity/2025 SE mechanism context	2/3 indirect	T1 recovery; T2 irreversible; T2* includes static dephasing	Medium
Slice selection, phase/frequency encoding	S4 44-60	2023 3(b)ii-iii; 2024 3(b)i; 2025 3(a)ii-iii,3(b)i	3/3	Full RF BW=(gamma/2pi)G delta-z; phase retains position information	High
GE timing, acquisition time, multislice	S4 61-74	2024 3(b)i-ii; 2025 3(b)i-iv	2/3	TR*Nphase*NEX; RF center changes between slices; gradients overlap correctly	High
Contrast, Ernst angle, SE, 3D, EPI, TSE	S4 75-86	2024 3(b)iv-v; 2025 3(b)v-vi	2/3	GRE steady state; optimum not always 90 degrees; SE refocuses static offsets	High / coverage
k-space, FOV, aliasing, SNR	S4 66-71,87-92	2025 scan-time application	1/3 time only	FOV=1/delta-k; finite kmax limits detail; voxel/SNR/time trade-off	Medium
MRI hardware, magnet types, coils	S4 93-101	2023 3(a)ii-iii	1/3	Main B0; three gradients; RF Tx/Rx; homogeneity, stability	Medium
MR angiography and fMRI	S4 102-108; syllabus p.1	No direct subquestion	0/3	Inflow contrast; BOLD haemodynamic proxy; EPI and task/rest correlation	Coverage
US physics, impedance, echo modes	S05 6-19	2023 4(a)ii; 2025 4(a)i	2/3	Z=rho*c; intensity reflection coefficient is pressure coefficient squared	High
TGC, piezoelectricity, arrays, focusing	S05 20-25,30-31,40	2024 4(a)iv (arrays)	1/3	Tx/Rx reciprocity, delays, focal beam width, TGC cannot recover lost echoes	Medium
US resolution, penetration, timing	S05 16-19,26-29	2023 4(a)iii-iv; 2024 4(a)iv-vi; 2025 4(a)iii-v	3/3	Round trip 2d; axial=Nc*c/(2f); frame=c/(2d*Nlines)	High
US compounding and artifacts	S05 32-38	2024 4(a)ii-iii; 2025 4(a)ii	2/3	Coherent speckle; angular/frequency compounding; shadow/enhancement/reverberation	High
Doppler	S05 39; syllabus p.1	No direct subquestion	0/3	Moving blood causes frequency shift; direction dependence; no detailed equation in slide	Coverage

Topic / subtopic	Slide reference	Exact historical evidence	Frequency	Understanding / equations / diagrams	Preparation priority
NM activity, half-life, radiotracers	S06 2-10	2023 1(c); 2024 1(b); 2025 1(b)iv	3/3	A=lambda*N; effective rate sums; activity vs atom count vs detected counts	High
Gamma camera, collimation, dead time, planar	S06 11-19	No direct equipment/dead-time subquestion	0/3	Direction selection, sensitivity penalty; nonparalyzable count correction	Coverage
SPECT, correction, SPECT/CT	S06 20-24	2024 1(b)i-ii	1/3	Rotating projections; scatter energy window; attenuation correction	Medium
PET, coincidence, PET/CT, detectors, 2D/3D	S06 25-40	2024 1(b)ii-iii; 2025 1(b)i-iii	2/3	511 keV pair, LOR, true/random/scattered events, functional/anatomical fusion	High
OCT principle, scans, resolution, applications, QA	Syllabus p.1; A2 PDF 3-8	2024 4(b); 2025 4(b)	2/3	Low-coherence interferometer; optical backscatter; A/B scans; shallow depth	High basics; assignment-only depth Medium
Thermal principle, applications, QA	Syllabus p.1; A1 PDF 3-9	2023 4(b)	1/3	Emitted IR to detector/ADC/calibrated surface temperature; environment and emissivity	Medium

Status: the syllabus explicitly lists all six major groups, including fMRI, Doppler, thermal and OCT. All supplied lecture topics are revision scope unless a later lecturer notice says otherwise. No verified exclusions were found. Details of OCT/thermal beyond the syllabus and historical questions are assignment-supported rather than confirmed lecture-level emphasis. CT QA is historically examined but lacks an explicit QA slide in the supplied main decks. The supplementary fMRI files and zipped ACA decks were not needed to establish scope and are not claimed as exhaustively reviewed sources.

7. Evidence-based preparation families

Confidence means strength of evidence for spending preparation time on a family, not the probability that an exact future question appears. The “additional coverage” family has low recurrence evidence but covers explicit syllabus content. No numerical probabilities are assigned.

1. MRI spatial encoding and sequence timing | High evidence

Representative prompt: Explain GE gradients; select a 4 mm slice; compute scan time; show multislice interleaving.

Evidence: 2023 3(b)ii-iii; 2024 3(b)i-ii; 2025 3(a)ii-iii,3(b)i-iv. Sources: S4 44-74. Three years of spatial encoding; diagrams recur in latest two.

Historical mark precedent: Typical slice calculation 3-4; GE/multislice 3-5 each; whole MRI block 25. Revise: Larmor Hz/rad conversion; full bandwidth; TR*Nphase; GE/multislice diagrams.

2. Ultrasound depth, resolution and frame rate | High evidence

Representative prompt: Choose the highest feasible frequency and determine frame rate at a stated imaging depth.

Evidence: 2023 4(a)iii-iv; 2024 4(a)iv-vi; 2025 4(a)iii-v. Sources: S05 6-19,26-29. Numerical family appears in all three papers.

Historical mark precedent: Depth/frequency 4-5; timing 2-4. Revise: 2*attenuation*f*d; c/(2d*Nlines); pulse length; focus.

3. Nuclear decay and count fractions | High evidence

Representative prompt: Find an activity equality time with physical/biological clearance or unequal initial atom numbers.

Evidence: 2023 1(c)i-ii; 2024 1(b)iv; 2025 1(b)iv. Sources: S06 6-8. A numerical decay task occurs every year with changing initial conditions.

Historical mark precedent: 5-6 per task; 10 total in 2023. Revise: A=lambda*N; effective half-life; logarithms; count efficiency assumptions.

4. X-ray attenuation, system and CT reconstruction | High evidence

Representative prompt: Explain radiography hardware; solve transmission/HVL; compare FBP and ART.

Evidence: 2023 2(a),2(b); 2024 2(a)iii,2(b); 2025 2(a),2(c). Sources: S2 7-42; S3 4-29. Complete 25-mark Q2 block every year; subtopic rotation prevents narrow selection.

Historical mark precedent: X-ray 11-14; CT 11-14 by year. Revise: Tube/grid diagram; exponential attenuation; Fourier-slice diagram; ART residual.

5. Image quality: PSF/LSF | High evidence

Representative prompt: Sketch convolution blur; derive FWHM and interpret resolving power.

Evidence: 2024 1(a)ii-iii; 2025 1(a). Sources: S1 39-48. Latest two years dedicate image-quality questions to spatial resolution.

Historical mark precedent: Definition 1; derivation 4; sketches 6; interpretation 3. Revise: Gaussian half-height algebra; delta and top-hat convolution; MTF.

6. PET/SPECT and radiotracer selection | High evidence

Representative prompt: Explain coincidence detection, PET/CT benefits and radiotracer properties.

Evidence: 2024 1(b)i-iii; 2025 1(b)i-iii. Sources: S06 9-10,20-34. Two recent papers use explanations alongside decay.

Historical mark precedent: 2-4 per conceptual part. Revise: 511 keV pair/LOR; collimator sensitivity; uptake; half-life.

7. US artifacts and compounding | High evidence

Representative prompt: Explain speckle, reverberation and posterior shadowing; propose angular/frequency compounding.

Evidence: 2024 4(a)ii-iii; 2025 4(a)ii. Sources: S05 32-38. Direct artifact emphasis in latest two papers.

Historical mark precedent: Speckle 3; three artifacts 6; mitigation 3. Revise: Interference vs random electronic noise; causal ray sketches.

8. OCT basics and application | High evidence

Representative prompt: Draw the optical paths and explain formation of A- and B-scans.

Evidence: 2024 4(b)i-ii; 2025 4(b)i-iii. Sources: Syllabus p.1; A2 PDF 3-7. Two recent optical blocks; no supplied lecturer OCT slides.

Historical mark precedent: 6-8 total OCT in past papers. Revise: Source/coupler/reference/sample/detector; path matching; retinal layers.

9. GRE Ernst angle and SE comparison | Medium evidence

Representative prompt: Determine optimum flip angle and percentage signal loss at 90 degrees.

Evidence: 2024 3(b)iii-v; 2025 3(b)v-vi. Sources: S4 22,78-81. Strong one-year numerical precedent plus recent sequence comparison.

Historical mark precedent: 2 angle + 7 SNR loss in 2024; comparison/mechanism 2+2 in 2025. Revise: Spoiled steady-state assumption; fixed TR/TE/noise; T2 vs T2*.

10. DSA, HU calibration, CT generations and QA | Medium evidence

Representative prompt: Derive log subtraction, calibrate HU, or compare rotating hardware.

Evidence: 2023 2(a),2(b)ii-iii; 2025 2(b). Sources: S2 43-49; S3 31-40. Each is evidenced, but alternatives rotate.

Historical mark precedent: DSA 11; generations 6; QA 4; HU calibration 4. Revise: Log sign; water/air reference; geometry; four QA checks.

11. ADC and thermal imaging | Medium evidence

Representative prompt: Explain quantization and thermal camera signal chain with QA controls.

Evidence: 2023 1(b),4(b). Sources: S1 62-72; A1 PDF 3-9. Absent in latest two but explicit syllabus and genuine precedent. Omission alone does not increase priority.

Historical mark precedent: ADC 8; thermal 8 in 2023. Revise: LSB, gain/clipping; IR optics to surface temperature; complementary use.

12. Additional syllabus coverage | Low evidence

Representative prompt: Explain fMRI BOLD, Doppler direction, k-space aliasing, gamma-camera dead time, dose or detector trade-offs.

Evidence: No direct dedicated question in these three papers. Sources: S4 66-71,87-108; S05 39; S06 11-17,35-40; S2 54-59. Protects against overfitting; scope is explicit, recurrence evidence is limited.

Historical mark precedent: No defensible historical mark estimate for these exact questions; model allocations are design choices. Revise: Basic mechanisms; supplied equations only; no advanced signal-analysis demands.

8. Source discrepancies and controls

• The historical sheet prints Q=dN/dt=lambda*N. Positive activity is A=-dN/dt=lambda*N. The model sheets correct the sign.

• The historical spin-echo intensity equation uses T2*. Ideal conventional spin-echo signal is proportional to rho*(1-exp(-TR/T1))*exp(-TE/T2). Static field-offset dephasing is refocused; intrinsic T2 relaxation remains. GRE retains T2* sensitivity. See S4 79-81 and Hornak, The Basics of MRI, chapter 10: https://www.cis.rit.edu/htbooks/mri/chap-10/chap-10-h5.htm .

• The 2025 slice question lacks B0, so absolute frequency endpoints cannot be numerically fixed; the bandwidth and endpoints relative to f0 can. Its stated 0.3 mT/cm is 0.03 T/m, not 0.0003 T/m. The models explicitly supply B0 when endpoints are required.

• The interleaved 2023 handwritten gradient conversion is unreliable: 0.5 mT/cm equals 0.05 T/m. A 3 mm full slab has delta-B=0.00015 T and full RF bandwidth about 6.39 kHz, not the much smaller handwritten value.

• 2024 “contrast” is undefined numerically: the lecture gives an absolute signal difference, while normalized contrast conventions differ. For the historical four-HVL bars, Ibar=I0/16; absolute gap-minus-bar signal is 15*I0/16. A normalized Weber contrast is 15/16; Michelson is 15/17. Models specify the convention.

• 2025 equal initial atom numbers imply unequal initial activities. A count-fraction calculation needs equal detection efficiencies (or supplied efficiencies). Models state this. The threshold wording “smaller than” means t must exceed the equality time.

• The local discussion Q4(a) has 400/8=50 Bq = 1.35e-6 mCi; the handwritten line omits the mCi conversion factor in its displayed denominator. Its population worked answer also has inconsistent powers of ten. Neither is an official marking key.

• S06 25 states a very large PET/SPECT SNR ratio. The models ask the physical sensitivity reasons without treating a fixed numerical ratio as universal. Likewise a universal requirement for an on-site cyclotron is too strong: frame the limitation as short-lived tracer production/distribution and cost.

• The supplied introductory slides simplify pixel meanings: MRI intensity depends on density, relaxation and sequence; B-mode US brightness is processed echo strength, influenced by impedance boundaries and scattering. MTF is the normalized magnitude of an optical transfer function, not generally a complex FT itself.

• Thermal/OCT sources are student reports. Basic OCT was cross-checked against the technical chapter cited by A2: https://www.ncbi.nlm.nih.gov/books/NBK554044/ . Technical model extensions are identified as assignment coverage, not proven lecturer emphasis.

9. Design decisions before model-paper generation

Paper 1 follows the strongest recent block pattern. Paper 2 rotates to DSA, hardware/phantom MRI and thermal imaging. Paper 3 increases application reasoning within supplied equations. Paper 4 retains four 25-mark blocks but broadens ADC, dose, detectors, fMRI, Doppler and QA coverage. Its allocations are revision design choices, not observed historical frequencies. Each is six pages: cover, corrected data sheet and one page per main question. Every numerical datum needed is supplied. Document C and the four individual solution PDFs contain proposed, non-official marking schemes.